# Complex Numbers: Polar Form, Powers, and Roots

Euler's Formula and De Moivre's Formula — 45 Minutes

Name: _____ Date: _____

## Instructions

Complete all 12 problems in 45 minutes without a calculator. You may use this reference page. Show polar or exponential setup when computing powers and roots. Give exact answers. Use rectangular form $a + bi$ unless another form is requested. In final polar or exponential answers, use positive modulus and angles in $[0, 2\pi)$. When finding roots, list every distinct solution.

## Reference Formulas

**Polar and exponential forms**
For $z = a + bi \neq 0$,

$$\begin{aligned} r = |z| &= \sqrt{a^2 + b^2} > 0, \\ a &= r\cos\theta, \qquad b = r\sin\theta, \\ z &= r(\cos\theta + i\sin\theta) = re^{i\theta}. \end{aligned}$$

Choose the angle using both coordinates and the correct quadrant. Arguments differing by $2\pi k$, $k \in \mathbb{Z}$, represent the same direction. Zero has modulus 0 and no defined argument.

**Euler's formula and Euler's identity**

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{i\pi} + 1 = 0$$

All angles are in radians. In particular, $e^{i(\theta+2\pi k)} = e^{i\theta}$.

**Products, quotients, and conjugates**
For $z_j = r_j e^{i\theta_j}$ with $r_j > 0$,

$$\begin{aligned} z_1 z_2 &= r_1 r_2 e^{i(\theta_1+\theta_2)}, \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}, \\ \overline{z} &= re^{-i\theta}, \qquad \frac{1}{z} = \frac{1}{r}e^{-i\theta}. \end{aligned}$$

**De Moivre's formula**
For $z = re^{i\theta} \neq 0$ and integer $m$,

$$\begin{aligned} z^m &= r^m e^{im\theta} \\ &= r^m\big(\cos(m\theta) + i\sin(m\theta)\big). \end{aligned}$$

A power multiplies the argument by $m$ and raises the modulus to the $m$th power. Negative powers require $z \neq 0$.

**All $n$th roots**
For $w = Re^{i\phi} \neq 0$ and positive integer $n$, the equation $z^n = w$ has exactly $n$ distinct solutions:

$$\boxed{z_k = R^{1/n} e^{\,i(\phi+2\pi k)/n}}$$

$$k = 0, 1, \ldots, n-1.$$

Equivalently,

$$z_k = R^{1/n}\left(\cos\frac{\phi + 2\pi k}{n} + i\sin\frac{\phi + 2\pi k}{n}\right).$$

The roots lie on the circle $|z| = R^{1/n}$ and are equally spaced by an angle of $2\pi/n$. For $n \geq 3$, they are the vertices of a regular $n$-gon.

The only distinct $n$th root of 0 is 0. To check any proposed root, raise it to the $n$th power and recover $w$.

## Complex Numbers: Polar Form, Powers, and Roots — Practice

1. Write each number in trigonometric polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$:

$$z = -3\sqrt{3} + 3i, \qquad w = -4 - 4i.$$

2. Evaluate exactly in rectangular form:

$$e^{i\pi} + 2e^{i\pi/2} - 3e^{-i\pi/3}.$$

3. Simplify and give the result in both exponential and rectangular form:

$$\frac{(2e^{5\pi i/6})(3e^{-\pi i/4})}{\sqrt{3}\, e^{\pi i/3}}.$$

## Complex Numbers — Continued

4. Express $\cos\theta$ and $\sin\theta$ using only $e^{i\theta}$, $e^{-i\theta}$, real constants, and $i$. Derive both formulas.

5. Evaluate in rectangular form:
$$(-\sqrt{3}+i)^9.$$

6. Evaluate in rectangular form:
$$\frac{(1+i)^8}{(\sqrt{3}-i)^6}.$$

## Complex Numbers — Continued

7. Let $z = 2e^{i\pi/12}$. Find both quantities in rectangular form:

$$z^{-6}, \qquad \frac{z}{\overline{z}}.$$

8. Find all cube roots of $-27i$. Give every root in exponential and rectangular form.

9. Find all fourth roots of $-8 + 8\sqrt{3}\,i$. Give every root in exponential and rectangular form.

## Complex Numbers — Continued

10. Find all fifth roots of $32e^{5\pi i/6}$ in exponential form. State the radius of the circle containing the roots and their angular spacing.

11. Find all complex solutions in rectangular form:

$$(z-1)^3 = 8i.$$

12. Find all complex solutions in exponential form:

$$z^6 + 8z^3 + 64 = 0.$$